QME STUDENT HANDBOOK

2013-2014

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Introduction

Quantitative methods in education (QME) consists of measurement, evaluation, and statistical techniques and models used in the study of educational phenomena. Graduates assist educational researchers in developing tests and surveys and other devices used to collect data in educational settings, analyzing these data, and evaluating educational programs. We are innovative in the development of new methods for analyzing education data, the creation of new assessment and evaluation models and methods, and new approaches to teaching and learning statistics. In many cases these activities focus on quantitative methodologies, but in others a mixture of quantitative and qualitative methods are used.

QME at the University of Minnesota is a track within the Educational Psychology graduate program in the College of Education and Human Development. The MA and PhD training in QME is designed for students to attain expertise in measurement, evaluation, statistics, and statistics education. Students who specialize in educational measurement study the theory and methods of measuring important variables in educational research, such as achievement, attitude, and specialized cognitive and non-cognitive constructs. They learn to develop tests and surveys for specific educational purposes, such as assessing the reading and mathematics achievement of elementary school students, and study issues related to measurement and accountability. Students specializing in educational evaluation learn to design effective approaches to evaluate the effectiveness of educational programs and education-related human services using a variety of quantitative and qualitative techniques, and may also study issues related to measurement and accountability. Students specializing in statistics develop an understanding of the relationship between research design and statistical analysis, and acquire skills in using a variety of statistical techniques appropriate for specific problems in education. Students specializing in statistics education investigate issues related to teaching and learning statistics and gain experience in statistics instruction. Skills in measurement, evaluation, and statistics prepare graduates for a variety of positions, including

test publishing firms, colleges, universities, research and evaluation centers, public school systems, state departments of instruction, and industry.

Because most QME students will obtain jobs in areas related to schools and education, students must possess a good knowledge of educational issues. For this reason, students are also required to take coursework outside QME.

Master's degree (MA) in educational psychology (emphasis in quantitative methods in education)

Purpose and goals

All students in the master's program are expected to develop the methodological competencies defined by the curriculum of the master's program (http://cehd.umn.edu/edpsych/Programs/QME/concentrations.html). Students successfully completing the degree are capable of participating in research of an applied nature, developing tests and surveys, translating research findings for application to educational settings, and assisting professional educators with measurement, evaluation, and statistical issues.

Doctoral degree (PhD) in educational psychology (emphasis in quantitative methods in education and concentration in measurement, evaluation, statistics, or statistical education available)

Doctoral studies are for students desiring advanced training in QME. Doctoral students develop methodological competencies defined by the doctoral curriculum (http://cehd.umn.edu/edpsych/Programs/QME/concentrations.html). Every effort is made to tailor the student's studies to their interests. For example, some students may be interested in attaining the skills to be a testing and measurement specialist, while others may wish to focus on developing the skills needed to conduct large scale evaluations, become a statistical data analyst, or to conduct research in statistical education (additional information on emphasizing statistical education appears in Appendix B). In each case students take specialized coursework in their area of interest and serve an internship in a related setting.

Students successfully completing the doctoral degree have an understanding of the content, methods, theories, and professional ethics associated with research methodology. They are capable of independently conducting research of an applied nature, translating sophisticated research findings for application in educational settings, and assisting educators with analyses and interpretations of research findings. Students also will have the skills to conduct methodological research, contributing directly to methodological practice and knowledge.

Research orientation of the program

The faculty strongly believes that collaborative and independent research activities should be central to a student's training and that these activities promote the acquisition of competencies essential to success upon attaining a degree. A premium is placed on involving students in research activities, including critically analyzing a body of literature, generating research questions which address specific issues in quantitative methods in education, such as developing empirically-based tests and surveys, designing and executing accountability studies in ways that take into account key methodological considerations (e.g., instrument construction and evaluation), correctly analyzing and interpreting research results, and succinctly communicating in writing the results of research studies. These competencies apply to both the master's and doctoral programs, but doctoral students are expected to develop the ability to work independently and demonstrate a greater depth of mastery of these competencies. Settings in which students frequently engage in research include collaboration with other students on course-related research, supervised research done under the direction of faculty researchers, and independently conducting a research study.

Studies for the master's (MA) degree

Student advisement

When a student is accepted for admission, a faculty member is assigned to be the student's academic adviser. The adviser acts as a liaison between the student and the faculty and is responsible for the following:

1. Supervising development of the student's degree program.
2. Approving all coursework at the time the student registers, including adds and drops.
3. Annually reviewing the student's academic progress and notifying the student if progress is unsatisfactory.
4. Certifying that graduation requirements have been met.

The student's adviser would normally supervise the student's master's research paper. The faculty member supervising the master's research paper is responsible for the following:

1. Assisting the student in identifying an appropriate research topic.
2. Supervising the development of the student's research paper.
3. Ensuring that the master's paper is prepared according to University and APA guidelines and is ready to be reviewed by other faculty.
4. Assigning a grade for the master's internship (EPsy 5272 or EPsy 8299).
5. Supervising the student's master's research paper (EPSY 5991) for Plan B.

How to progress through the program

Students should read the *University of Minnesota Graduate School Catalog* for details about master's degree requirements, which is available online at http://www.catalogs.umn.edu/grad/, and the *Educational Psychology Graduate Program Handbook* at www.cehd.umn.edu/EdPsych/handbook/.

1. Length of program

The MA program in Educational Psychology with an emphasis in QME requires successfully completing QME and Educational Psychology core courses, an internship, and a research paper (typically a Plan B paper). With normal progress, most students should be able to complete the degree in 4 terms of full-time study (see curriculum).

2. GPA

The Graduate School requires students to maintain a grade point average (GPA) of at least 2.80 on a 4.0 scale (A = 4.00, B = 3.00, C = 2.00, D = 1.00, and F = 0.0). Grades of C- or better and S (Satisfactory) are acceptable on the official degree plan, but grades of S are not calculated in the GPA. At least two-thirds of the course credits included on any degree plan form must be taken A-F. All EPSY core courses must be taken A-F as well.

3. Time Limit for Degree Completion

The Graduate School requires all master's work be completed within a period of five years after initial enrollment in the program.

4. Registration

Students must register before the beginning of a term to avoid late fees. To maintain their active status, graduate students must register every fall and spring semester. Those who do not register in the Graduate School each semester are considered to have withdrawn and their Graduate School records are deactivated. Inactive students may not register for courses, take examinations, submit a degree plan form, file for graduation, or otherwise participate in the University community as Graduate School students. Those who wish to resume graduate work must request readmission to the Graduate School and the program and, if readmitted, must register in the Graduate School for the term of readmission to regain their active status. Students who are readmitted will be responsible for the course and exam requirements in effect at the time of the readmission.

Faculty in QME are often not available during the summer term, and students should plan to complete their summer and fall term registrations before the end of the spring term.

5. Degree plan and committee

Students are required to file an official Degree Plan Form with the director of graduate studies (DGS) in the Department of Educational Psychology after

completing 10 credits (ordinarily no later than the second semester of the first year). This requires completing the Program Checklist and the Graduate Degree Plan form (www.grad.umn.edu/current_students/forms/gs89a.pdf) Students should list all coursework, completed and proposed, that will be offered in fulfillment of degree requirements, including transfer work (see "Transfer of credits and course waivers" below). Revisions to a degree plan can be made with the adviser's approval by submitting a Graduate School Petition form (www.grad.umn.edu/Current_Students/forms/masters/index.html) to the DGS. After the degree plan has been approved assign members to your master's final examination committee at http://www.grad.umn.edu/students/forms/masters/index.html.

6. Transfer of credits and course waivers

Students may transfer up to 14 credits (or up to 40%) of graduate coursework taken either at another institution or at the University of Minnesota prior to admission may be used towards a master's program if approved by the program faculty and the Graduate School. (Courses must be taken at the graduate level and appear on a graduate transcript). Students who wish to have courses waived because of previous coursework, experience, or other proof of competence must complete the Department of Educational Psychology Internal Petition Form. Approval of a course waiver does not change the total number of credits a student must earn in research methodology for the master's degree. The Internal Petition Form for waiving a course is available at www.cehd.umn.edu/EdPsych/forms/.

7. Educational psychology core requirements

Students must satisfy the educational psychology core requirements of at least 3 credits in learning/cognition or social psychology/personality, 3 credits in statistics, and 3 credits in measurement. Students must also take 3 credits of EPSY 5991 or EPSY 8994 (Plan B) or 10 thesis credits, EPsy 8777 (Plan A). All EPSY core courses must be taken A-F.

8. QME core requirements

Information about required QME courses is available at the curriculum website. Descriptions of all QME courses are available online.

9. Master's research paper (Plan B)

3 credits of EPSY 5991

Students demonstrate their knowledge of research and scholarship by working independently on a research paper involving a minimum of 120 hours of work. Typically this paper or project resembles a thesis in volume; however, it differs from a thesis in that it need not represent an original, empirical study. For example, a student may prepare a library-based research paper that reviews, organizes, synthesizes, and critiques previously conducted research and scholarly discussion pertaining to issues in measurement and accountability or statistics/statistical education. A paper that involves the analysis or re-analysis of previously collected data may also satisfy this requirement. Two faculty members from quantitative methods in education and one faculty member with graduate faculty membership outside of educational psychology serve as readers. The paper is completed in three stages. First, a 1-3 page description of the topic of the paper and proposed methods is presented to the readers for approval. Second, the paper is completed under the adviser's supervision and presented to the readers for approval. Third, the student defends the paper in a final oral exam. Students should check with their adviser for details concerning the oral exam. A suggested format for the master's paper appears in Appendix A. Assuming other program requirements have been met, students may work on their research paper concurrent with serving their internship.

The Graduate School allows for a Plan A option – under agreement with your adviser this may be an option for students with significant prior experience and wishing to take fewer courses. The Plan A requires registration for 10 Thesis credits (EPSY 8777).

10. Graduation requirements

Students should obtain a Graduation Packet early in the term they wish to graduate (www.grad.umn.edu/current_students/masters/index.html). This packet has two forms that must be completed (a) Application For Degree and (b) Thesis Reviewers Report Form—Plan A; Final Examination Report—Plan B. The Application for Degree must be submitted to the Office of the Registrar (200 Fraser Hall) by the first working day of the intended month of graduation. The Final Examination Report form must be in the Graduate School by the last working day of the intended month of graduation.

Studies for the doctoral degree (PhD)

Student advisement

When a student is accepted for admission, a faculty member is assigned to be the student's academic adviser. The adviser acts as a liaison between the student and the faculty and is responsible for the following:

1. Supervising development of the student's degree plan.
2. Approving all coursework at the time the student registers, including adds and drops.
3. Annually reviewing the student's academic progress and notifying the student if progress is unsatisfactory.
4. Assisting the student in identifying an appropriate pre-dissertation research project, specialty prelim exam, preliminary oral, and dissertation topics.
5. Certifying that graduation requirements have been met.

The student's adviser also supervises the student's dissertation research. The faculty member supervising the dissertation is responsible for the following:

1. Assisting the student in identifying an appropriate research topic.
2. Supervising the development of the student's dissertation.
3. Ensuring that the dissertation is prepared according to University and APA guidelines and is ready to be reviewed by other faculty.
4. Assigning a grade for internship (EPSY 5272 or EPsy 8299).

How to progress through the program

Students should read the *University of Minnesota Graduate School Catalog* for details about doctoral degree requirements, which is available online at www.catalogs.umn.edu/grad/, and the *Educational Psychology Graduate Program Handbook* at www.cehd.umn.edu/EdPsych/handbook/.

1. Length of program

The PhD in Educational Psychology with a QME focus requires successfully completing QME and Educational Psychology core courses, and 24 thesis credits, a pre-dissertation research paper, written specialty prelim, preliminary oral examination, and a dissertation. Students with a master's degree in QME can count 35 master's credits toward a doctorate in QME. (Students who emphasized statistical education in their master's work can count 41 credits toward the

doctorate.) Students already possessing a master's in QME may be able to complete the degree in 2 years of full-time study. Students entering the program with a master's degree in a related field may be able to complete the degree in 3 years of additional full-time study. Students entering the doctoral program without a master's degree would normally complete the program in 5 years of full-time study.

2. GPA

The Graduate School requires students to maintain a grade point average (GPA) of at least 3.00 on a 4.0 scale (A = 4.00, B = 3.00, C = 2.00, D = 1.00, and F = 0.0). Courses with grades of A, B, C (including C-), and S may be included in the official degree plan, but grades of S are not calculated in the GPA. Students pursuing a doctoral degree must register for doctoral thesis credits (EPSY 8888); these registrations are not graded and therefore cannot be used to meet course credit requirements. At least two-thirds of the total number of course credits included in any degree plan and all EPSY core requirements must be taken A-F.

3. Time Limit for Degree Completion

The Graduate School requires all doctoral work be completed within a period of eight years after initial enrollment in the program.

4. Registration

Students must register before the beginning of a term to avoid late fees. To maintain their active status, graduate students must register every fall and spring semester. Those who do not register in the Graduate School each semester are considered to have withdrawn and their Graduate School records are deactivated. Inactive students may not register for courses, take examinations, submit degree plan or thesis proposal forms, file for graduation, or otherwise participate in the University community as Graduate School students. Those who wish to resume graduate work must request readmission to the Graduate School and the program and, if readmitted, must register in the Graduate School for the term of readmission to regain their active status.

Students who are readmitted will be responsible for the course and exam requirements in effect as of the readmission term.

Faculty in QME are often not available during the summer term, and students should plan to complete their summer and fall term registrations before the end of the spring term.

5. Degree plan

Students who have completed four semesters of study in the doctoral program must file an official Degree Plan Form with the director of graduate studies (DGS) in educational psychology. The Educational Psychology Department requires doctoral students to submit their degree plan forms to the DGS prior to taking the Specialty Written Prelim Exam. This should be by the end of the second year of graduate study, and at least one semester prior to the term during which the Prelim Oral Exam will be taken. This requires completing the Program Checklist, and the Graduate Degree Plan form (www.grad.umn.edu/current_students/forms/gs89a.pdf. Students should list all coursework, completed and proposed, that will be offered in fulfillment of degree requirements, including transfer work (see Transfer of credits and course waivers below). Revisions to a degree plan can be made with the adviser's approval by submitting a petition to the director of graduate studies. Revisions to a degree plan can be made with the adviser's approval by submitting a Graduate School Petition form (www.grad.umn.edu/Current_Students/forms/doctoral/index.html) to the DGS. After the degree plan has been approved, assign members to your preliminary oral examination committee at http://www.grad.umn.edu/students/forms/doctoral/index.html. QME students may not use QME faculty members with joint graduate school appointments or external affiliations to fulfill the external membership requirement on graduate committees.

6. Transfer of credits and course waivers

There is no limit on the number of graduate credits obtained at another institution that can be transferred, but any doctoral credits a student wishes to transfer must be approved by the QME faculty and the Graduate School. See the *Graduate School Catalog* (http://www.catalogs.umn.edu/grad/gen/phd.html) for details about transfer work or coursework taken at the University of Minnesota prior to admission.

Students who wish to have courses waived because of previous coursework, experience, or other proof of competence must complete the Department of Educational Psychology Internal Petition Form. This form is available at www.cehd.umn.edu/EdPsych/forms.

7. Educational psychology core requirements

Students must successfully complete 9 credits from at least two areas of history/systems, learning/cognition, orsocial psychology/personality. QME students must complete 9 credits in research methods (EPSY 8215, 5244 and 5247). Students must also take 9 elective credits in EPSY (QME core courses may be used to satisfy this requirement) and 24 EPSY 8888 thesis credits. All EPSY core courses must be taken A-F.

8. QME core requirements

Information about required QME courses is available at the curriculum website. Descriptions of all QME courses are available online.

9. QME Doctoral Written Preliminary Exam Requirements

There are three requirements for the doctoral written prelim exam:

a. Courses

QME doctoral students take three research methods core courses: Survey Design, Sampling, and Implementation (EPSY 5244), Qualitative Research Methods (EPSY 5247), and Advanced Research Methods (EPSY 8215).

b. Pre-Dissertation Research Project

Students must successfully complete a pre-dissertation research project under the supervision of their adviser, which entails a research review, research proposal, study implementation, and writing up of the results. A student with a master's degree in QME automatically satisfies the pre-dissertation research paper requirement. Students who need to satisfy this requirement should begin work on their research review in their second term of doctoral study, develop a research proposal in their third term, and plan to conduct the study and write up the results during their fourth term of doctoral study. The paper may be a synthesis of existing work, pilot study, and/or may be based on existing data. The resulting pre-dissertation research paper would normally have a structure similar to the outline for the master's paper in Appendix A and should not exceed 25 pages. The paper is read by the student's adviser and one other faculty member in QME. The pre-dissertation project can be done in conjunction with EPSY 8215. Additional information about the pre-dissertation research requirement is available in the Educational Psychology Graduate Program Handbook. Submit the QME Predissertation Research Project form to the program staff in 250 EdSciB.

c. Specialty Preliminary Exam

Purpose of the specialty prelim exam

The purpose of the specialty prelim exam is to assess the student's mastery of measurement, evaluation, and statistics and the student's acquisition of both

breadth and depth of understanding of the general field of quantitative methods in education.

Eligibility for the specialty prelim exam

In order to sit for the QME specialty prelim exam the student must have a Degree Plan on file. In addition, at least 6 of the 9 EPsy Research Methods and QME Core courses and the Predissertation Research Project must be completed before writing the paper. Students must pass Part I (if required) before beginning work on Part II.

Scheduling the written specialty prelim

Registration for the specialty prelim is done through the student's adviser and the director of graduate studies. The EPSY Prelim Registration Form is available at http://www.cehd.umn.edu/EdPsych/Current/QME.html.

Description of the specialty prelim exam

The examination has two parts. In Part I, students sit for a constructed-response exam covering measurement, evaluation, and statistics. In Part II, the student prepares a potentially publishable paper. Students who are in good standing following the annual review at the end of their third year will be exempt from Part I of the speicialty prelim exam. Students who are NOT in good standing following the annual review at the end of their third year must complete Parts I and II of the specialty prelim exam. To be considered in good standing, students must have a degree plan on file, a GPA of at least 3.50 in the core QME and research methods courses (at least 6 courses completed) and have completed the predissertation research project prior to spring term of their third year.

All QME students must complete Part II of the Doctoral Written Preliminary Examination, with the only exception being a student who has already written and possibly published an appropriate research paper in the field of QME. In this case, the student should petition their adviser to request that the already-written research paper be substituted for the Part II requirement. Note that the research paper for Part II is a separate requirement from the pre-dissertation research project.

Part I. Sit-down examination

The student sits for five questions, three in their major area (measurement, evaluation, or statistics), and one each in the remaining areas (measurement, evaluation, or statistics). Students will be given four questions in their major area and will select three to respond to; for the remaining areas students will be given two questions in each area and asked to select one to respond to. These questions will be of the constructed-response type, and will be prepared by the student's Doctoral Prelim Committee. This committee will consist of three members of the QME faculty. Students will have four hours to answer the five questions (approximately 45 minutes each). Students may not bring any materials to the exam, and will type their responses into a word-processing program on a computer that will be provided. Student responses will be read by members of the Doctoral Prelim Committee, who will make a pass/fail recommendation to the full QME faculty. Students have two opportunities to pass the written exam. If the student fails the exam both times the student's study in the QME/Ed Psych program is terminated.

Students sitting for Part I are responsible for the material in the required doctoral courses, including, but not limited to, course notes, textbooks, journal articles, and technical reports used in these classes. The exams will be tailored to the students taking the exam at a given time, given their interests and coursework histories. Readings that should be consulted during preparation for the written exam include materials provided in courses—each area has identified a core set of readings including seminal articles, comprehensive texts, and books with a specific focus. These lists are available from faculty and can also be found at http://www.cehd.umn.edu/EdPsych/Programs/QME/prelimquestions.html.

Part II. Student submits a potentially publishable research paper

In conjunction with their adviser or another faculty member in QME, the student proposes a topic for the research paper. The topic of the paper should be agreed to by the student and their adviser before the student begins work on the paper. Students may use their work from EPSY 8215 and/or their pre-dissertation

research project as a basis for the research paper. However, the research paper cannot be limited to work in EPSY 8215 or the pre-dissertation research project; rather, the paper must be sufficiently different from other research papers the student has written to merit treatment as a new piece of work that will satisfy the doctoral written specialty exam requirement. In general, the structure of the paper will follow that outlined for the master's research paper. Data for the paper may be collected by the student as part of satisfying this requirement or may come from another source, for example, data from a research project or an extant dataset in the public domain. There are no specific requirements for the length of the paper, but the paper should follow APA conventions. As a guideline, students should plan to complete the research paper within one semester. Once completed, students are strongly encouraged to submit their work for possible publication in an appropriate journal.

Specialty prelim exam scoring and required forms

The evaluation of exam responses will be based on the following principles: (a) Does the response constitute a logical argument exhibiting an awareness of the depth and breadth of issues related to the question? (b) Does the response address a research base in responding to the question? (c) Does the response demonstrate professionalism? These principles will be considered as well as the accuracy and completeness of all responses.

The Part I exam will be scored by at least two QME faculty. Raters will score exam questions on a scale of 1-5 with 5 the highest possible score; scores 2.5 and above are passing. The scores for each rater will then be averaged for an overall final score. If the average total score is below 2.5, the student fails. If there are major discrepancies between the original raters, additional raters may be asked to score the exam. Students can sit for the Part I exam twice. Students will normally be informed of their mark on the Part I exam within one month of completing the exam.

The Part II research paper will be read by at least two QME faculty, one of whom will be the student's adviser. One review of the final draft of the paper by the

adviser is allowed prior to submission for 'scoring'. The readers will make a pass/fail recommendation to the QME faculty. Students will normally be informed of their mark on the Part II paper within one month of completing the exam. Students who fail Part II have one additional opportunity to pass this portion of the doctoral prelims. Repeating Part II may entail substantial revision of the original paper or may require the writing of a new paper under the supervision of the adviser.

The entire exam (Part I if required and Part II) can be retaken once. If needed, either part of the exam can be retaken. If the student fails Part I twice or is not able to revise and successfully pass Part II, the student's enrollment in the QME track in the Educational Psychology Department is terminated.

If a scheduled prelim is missed and not taken as scheduled, the student must petition the adviser or prelim committee for permission to take the prelim on a new date. Petitions will be reviewed on an individual basis. Typically, petitions are accepted only from those students facing extraordinary circumstances. If a petition is denied, the missed prelim will be recorded as a fail.

Upon successful completion of Part I and/or Part II of the specialty exam, the QME Written Specialty Exam Completion Form, available at http://www.cehd.umn.edu/EdPsych/Programs/QME/WritPrelim_completion.pdf, should be submitted to the program staff in 250 Education Sciences Building.

When three requirements of the written prelim exam are completed (coursework, pre-dissertation research project and the specialty exam), submit the departmental checklist for showing completion of the written prelim (http://www.cehd.umn.edu/EdPsych/forms/default.html) to the DGS Assistant in 250 Education Science Building.

10. Preliminary oral examination

After successfully completing the doctoral written preliminary exam, students must pass a preliminary oral exam that involves writing a paper that integrates the literature in a proposed research area. In many cases, the preliminary oral

paper will be the basis of the Review of the Literature chapter in the dissertation. The length and content of the paper are negotiated by the student with their adviser, but generally no more than 30 pages. With the adviser's approval, the paper is presented to the student's Examining Committee for review. The Examining Committee consists of at least three QME faculty and one faculty member on the graduate faculty with an appointment outside the Department of Educational Psychology. QME students may not use QME faculty members with joint graduate school appointments or external affiliations to fulfill the external membership requirement on graduate committees. Committee members are identified after the Degree Plan is filed. After the degree plan has been approved assign members to your preliminary oral examination committee at http://www.grad.umn.edu/students/forms/doctoral/index.html. Typically the faculty serving on a student's Examining Committee for their preliminary oral also serve on their doctoral committee. Passing the preliminary oral exam allows the student to begin work on their dissertation and register for thesis credits. Students who fail the oral preliminary exam have one additional opportunity. Attempting to pass the preliminary oral exam a second time may entail substantial revision of the original preliminary oral paper or may require the writing of a new paper.

The prelim oral exam must be scheduled with the Graduate School at least one week prior to the exam. Additional information on the preliminary oral examination can be found at www.catalogs.umn.edu/grad/gen/phd.html and in the *Educational Psychology Graduate Program Handbook*.

11. Doctoral dissertation fellowship

The Graduate School awards doctoral dissertation fellowships on a competitive basis to support doctoral students during the dissertation writing stage of their graduate career. Students must be nominated by their program. Competition for these awards is fierce but we do encourage all eligible students to apply. Educational Psychology students have been successful in receiving this fellowship. To be eligible for the award, students must have completed all

coursework on their degree plans, passed the written and oral prelim exams, and have submitted the thesis planning panel form.

Early in the spring semester, students will be asked to submit proposals to the program. The Graduate Advisory Committee will review the proposals and submit the most promising to the Graduate School.

Information on the doctoral dissertation and other fellowships is available at http://www.grad.umn.edu/fellowships.index.html.

12. Doctoral dissertation

Each student writes a dissertation that presents the results of the student's dissertation research. An appropriate dissertation research project involves significant, original, and independent research work that is grounded in a body of literature. It presents hypotheses tested by data and analyses and provides a contribution or advancement in the fields of quantitative methods in education. It is the responsibility of the student's doctoral committee to evaluate the dissertation in these terms and to recommend awarding the doctorate only if the dissertation is judged to demonstrate these qualities. Assign members to your final oral examination committee at http://www.grad.umn.edu/students/forms/doctoral/index.html. Additional information on the dissertation is available in the *Educational Psychology Graduate Program Handbook* and online at www.catalogs.umn.edu/grad/gen/phd.html.

The dissertation has three phases (a) prospectus phase, (b) research phase, and (c) defense phase.

(a) In the prospectus phase students work under the supervision of their adviser to prepare a prospectus that describes the proposed research. This document is presented to the student's Thesis Planning Panel, which include the student's adviser and two other faculty who normally are members of the student's doctoral committee. After approval by the Thesis Planning Panel, the student submits the Thesis/Project Proposal form to the director of graduate studies for

approval. (b) The student carries out the proposed research under the adviser's supervision and prepares the written dissertation. The dissertation is then submitted to the Thesis Planning Panel. If they agree that the dissertation is ready to defend, the student submits the Thesis Planning Panel form available at http://www.cehd.umn.edu/EdPsych/forms/default.html to the director of graduate studies assistant. (c) In the defense phase, the student defends the written dissertation to their doctoral committee in a final oral defense. Additional information on the final oral is available in the *Educational Psychology Graduate Program Handbook* and www.catalogs.umn.edu/grad/gen/phd.html.

13. Graduation requirements

Students should obtain a Graduation Packet prior to the term they wish to graduate ((www.grad.mn.edu/students/doctoral/index.html). This packet has two forms that must be completed: (a) Application For Degree and (b) Reviewers Report. The Application for Degree must be submitted to the Office of the Registrar (200 Fraser) by the first working day of the intended month of graduation. The Reviewers Report form must be returned to the Graduate School at least one week prior to the scheduled date of the final oral examination.

The signed Final Oral Examination Report form must be returned to the Graduate School no later than one working day following completion of the Final Oral Defense. All other materials (copy of the dissertation abstract, a signed copy of the dissertation, the Microfilm Agreement Form and the Survey of Earned Doctorate) must be submitted to the Graduate School by the last working day of the intended month of graduation.

Appendix A

Sample format for the MA and PhD paper

This is the format for the PhD paper, however this outline is offered as one suggestion for how to organize a master's research paper as well. The major difference would be that the chapter titles become the section headers in the MA thesis. Other outlines may be appropriate also. Check with your adviser for advice on how to organize your particular paper.

Title page

Title of the work, names of the author, degree(s) of the author, department, and university to which it is submitted, date it is submitted.

Acknowledgments

Acknowledge all academic and technical assistance you obtained at each stage of the study.

Abstract

500 word abstract summarizing the study and its findings.

Table of contents

List all preliminary sections, all chapter titles, all major headings and subheadings of each chapter, reference list, and appendices. Include list of tables and list of figures. Page number of the beginning of each section (subsection) listed is put in the right hand margin.

Chapter I. Introduction

1. General background introduction
2. Purpose of the study, rationale, theory
3. Statement of the problem (i.e., hypothesis to be tested or research questions to be investigated)
4. Limitations of the study
5. Definitions of special terms you will use

Chapter II. Review of the literature

1. Organize and synthesize previously conducted research and scholarly discussion related to the problem being investigated. The goal of this section is to show how each of your hypotheses or research questions are derived from the previous work of others.

2. End chapter with a summary section

Chapter III. Method

Write this chapter before conducting study and present it along with Chapters I and II as the research proposal. After the data analyses are complete, revise this chapter.

1. Sample
2. Procedures and research designs
3. Instruments used/developed and their reliability/validity
4. How each research question/hypothesis will be tested
5. State hypothesis
6. What analysis research design will be carried out?
7. How will the results be organized and presented?

Chapter IV. Results and analysis

1. General data analysis and results
2. Data results specific to each hypothesis are presented
3. Chapter review

Chapter V. Discussion, interpretation, conclusion, summary

Summarize the entire project including what hypothesis/questions were investigated, why they were investigated, how they were investigated, the major findings, and your conclusions.

1. Discuss the findings and the hypothesis in a holistic and integrated fashion.

2. Explain any extraneous factors that may have led to the results you obtained.

3. Discuss the practical and theoretical implications of your findings and precisely how your research supports each implication.

4. State the conclusions to be drawn from your entire study (including review of the literature and empirical findings; i.e., integrate everything).

5. Discuss suggestion for future research, next stages of research, what others might do to follow up on your study.

References

List every reference cited in the body of the paper in alphabetical order by last name of author. Use APA style for each entry. List only those references actually cited in the body of the paper.

Appendices

Include long data tables, copy of instrument(s) used, special computer program written especially for this study, supplementary illustrated material, letters soliciting subjects, consent forms, and other material that supplement the text but which are not appropriate for inclusion in the text.

Appendix B

Information for students with concentrations in statistical education in the master's or doctoral quantitative methods in education (QME) area, or for non-QME students attaining a PhD minor in educational psychology with an emphasis in statistical education

Introduction

The number of courses in introductory and intermediate statistics is steadily growing. At the college level, there are now more sections of introductory statistics than sections of calculus. At the high school level, there are over 40,000 students in Advanced Placement statistics courses, and those numbers are steadily rising. Despite the increase in statistics courses, there is currently no formal preparation for the teaching of statistics. Traditionally statistics teachers at the high school level have bachelors or master's degrees in mathematics or mathematics education, and may not have ever taken an applied statistics course, worked with data, or used a statistical software package. At the college level, most teachers of statistics are mathematicians although some are statisticians. Neither graduate discipline prepares their students to be teachers of statistics.

Over the past two decades much attention has been paid to first courses in statistical science. Calls for reform have made recommendations about how the teaching of these courses should be improved to increase student learning and attitudes towards statistics, and to develop a statistically literate society. This reform movement has greatly affected the teaching of the introductory statistics course, updating the content, placing more of an emphasis on data analysis using real data and simulation, including material on designing experiments, sampling, and surveys as ways of collecting and producing data, and incorporating the use of technology (both software and Web resources) as an integral part of the course.

There is also a need to attract and prepare more people to the field of statistics, which has led to a major initiative of the American Statistical Association: The

Undergraduate Statistics Education Initiative. One focus of this initiative is the first course in statistics, and one recommendation from the working group on this topic is to develop programs to better prepare teachers of statistics. Second, intermediate courses in statistics are also beginning to attract more attention and National Science Foundation funding is now going to the development of improved and innovative second courses in statistics.

The master's and doctoral degree in educational psychology at the University of Minnesota is the first in the U.S. to offer a specialization in statistics education.

Goal of the master's and doctoral program emphasis in statistical education. All students in the master's and doctoral programs are expected to develop their knowledge of areas related to statistics education as well as methodological competencies defined by the curriculum of the programs. A teaching internship will allow students to apply what they learn to the classroom setting and receive supervision and feedback on their teaching. The research requirement for the master's thesis or doctoral dissertation will prepare the students to conduct high quality educational research applied to the teaching and learning of statistics, and to link the results of research to classroom practice.

Examples of employment opportunities for students graduating with an emphasis in statistical education. Students entering the master's program in QME (emphasis in statistics education) at the University of Minnesota will be prepared to teach introductory and intermediate statistics courses in a high school or at a community or professional college. (Please note that this is not a K-12 licensure program.) Students entering the doctoral program will be prepared to teach statistics at the college level and to conduct research on the teaching and learning of statistics.

Plan of studies for students with a concentration in statistical education. Master's students must complete a minimum of 41 credits, including 19 credits in measurement, evaluation, and statistics, 3 credits in learning and cognition, and 3 credits in social psychology or personality. Students must also take two courses in statistics education and at least 6 credits outside of the Department of

Educational Psychology. These courses may come from math education and/or statistics. Information about required courses for master's and doctoral students with a concentration in statistical education is available Areas of concentration.

Doctoral students emphasizing statistical education must also spend one semester (3 credits) in an internship. The goal of the internship is to apply knowledge of research on how students learn statistics and awareness of current resources and recommendations for teaching first and second courses in statistics. Students would experience supervised teaching, at the high school or college level. The students may arrange this for themselves or may request help in arranging a job placement. One option is to teach one of the several undergraduate statistics courses in the department. Each semester all students involved in the teaching internship would meet together with a supervising faculty member on a regular basis to discuss teaching issues. The faculty member would also observe some classes and provide feedback and suggestions regarding the students' teaching.

PhD minor educational psychology with an emphasis in statistics education (12 credits). Designed for PhD students outside educational psychology and quantitative methods in education who desire specialized training in statistical education. Prerequisites are coursework in statistics equivalent to EPSY 8261, EPSY 8262, EPSY 8263, and EPSY 8264. Students must take EPSY 5271—Becoming a Teacher of Statistics, and EPSY 8271—Statistics Education Research Seminar: Studies in Teaching and Learning Statistics, one course in learning and cognition, and spend a semester in a teaching internship, EPSY 5272.